DERIVATIVES

Rolle's Theorem

Click a hypothesis to see what it does and what breaks without it.

IF

1. f is continuous on $[a, b]$
   CLICK: DROP IT
2. f is differentiable on (a, b)
   CLICK: DROP IT
3. $f(a) = f(b)$
   CLICK: DROP IT

$\Longrightarrow$

THEN

$$f'(c) = 0 \text{ for some } c \in (a, b)$$

A function that starts and ends at the same value must have a horizontal tangent somewhere between.